

CIRCULAR CARBON ECONOMY

Since the advent of fossil fuels, Carbon followed a linear path: extraction, conversion to energy and release of CO₂ and other GHG emissions into the atmosphere. This model powered industrialisation, global trade and rising standards of living. It also led to increases in atmospheric stock of CO₂ and other GHG emissions, with consequential adverse impact of Climate Change

Circular Carbon Economy is a framework for carbon cycle, under circular economy framework, viz Reduce Reuse, Recycle and Remove. Rather than relying solely on eliminating emissions the traditional "linear" model

DEFINITION OF CIRCULAR CARBON ECONOMY (CCE) : as per the UNFCCC Circular Carbon Economy page” is summarized below.

Climate change mitigation focused mainly on faster deployment of renewables and replacement of hydrocarbons in the power and transportation sectors. However, global emissions are still growing and greenhouse gas (GHG) concentrations are still at high levels. It seems the current approach is not working.

Mitigation strategies should consider practical and effective solutions that create a holistic approach and make use of all the opportunities for solving this issue. This approach would accommodate a diverse energy mix of renewable and traditional fuel sources, would lessen atmospheric levels of GHGs, while also sustaining economic growth and prosperity.

As important as renewables are, and as much progress as they have made in recent years, few analysts believe that the world can achieve the Paris Agreement’s goal of a balance between sources and sinks by 2050 through renewables alone. The global path toward a carbon balance will inevitably include fossil fuels, and their carbon emissions must be managed.

The concept of the circular carbon economy, an outgrowth of the idea of the circular economy, is a useful framework for understanding how all carbon mitigation options can be linked together in a system that achieves the climate goals laid out in the Paris Agreement.

CCE OFFERS A NEW WAY OF APPROACHING CLIMATE GOALS that implicitly values all options and encourages all efforts to mitigate carbon accumulation in the atmosphere. The circular carbon economy extends the concept of a circular economy (reduce, reuse, recycle) by including remove and focusing exclusively on carbon and energy flows.

CCE is a framework that manages carbon the way a traditional economy manages resources. Instead of treating carbon as just a waste product or a one-time resource, the CCE follows four core strategies (known as the "4 Rs") to close the loop on emissions:

Reduce: Cut down the amount of carbon emitted in the first place through energy efficiency and renewables.

Reuse: Capture emissions and put them directly to use without breaking them down, such as in enhanced oil recovery or certain manufacturing processes.

Recycle: Transform waste carbon dioxide CO₂ into valuable products like synthetic fuels, concrete, plastics, or fertilizers.

Remove: Pull carbon directly out of the air using nature-based solutions (like planting trees) or engineered technologies, then store it underground.

SPECIFIC TECHNOLOGIES IN THE CIRCULAR CARBON ECONOMY (CCE) that drive the transformation of carbon from an environmental liability into an industrial asset. These technologies align with the framework's core pillars:

1. **Reduce (Efficiency & Substitution):**

Direct Reduced Iron (DRI): Uses green hydrogen or natural gas instead of coal to extract iron from ore, cutting steel industry emissions.

Allam Cycle Power Plants: An advanced oxy-fuel combustion process that burns natural gas with pure oxygen, capturing 100% of the produced carbon dioxide (CO₂) inherently as part of the generation process. The Allam Cycle uses high-pressure, supercritical carbon dioxide (sCO₂) as the fluid that spins the turbines to generate electricity. After spinning the turbine, most of the CO₂ is compressed, reheated, and sent right back to the combustor to keep the cycle going.

2. Reuse (Direct Utilization):

Enhanced Oil Recovery (EOR): Injects captured, compressed CO₂ gas into maturing oil fields. The gas swells the trapped oil, making it easier to extract while trapping the CO₂ underground.

Supercritical CO₂ Working Fluids: Uses CO₂ in a high-pressure, semi-liquid state to replace standard refrigerants in industrial cooling systems.

3. Recycle (Conversion to Products)

In-Situ Mineralization: CO₂ in flue gases (Hydrogen generators, Gas Engines, Cement Kilns, Steel Plants) is mineralized to useful chemical compounds, which leads to permanent storage of CO₂ (eg CaCO₃)

Ex-Situ Mineralization: Reacts captured CO₂ with industrial waste (eg steel slag or fly ash) to produce synthetic concrete aggregates and solid building blocks, permanently trapping the gas.

Power-to-X (P2X) / Synthetic Fuels: Combines captured CO₂ with green hydrogen via a catalyst to synthesize e-methanol, methane, and Sustainable Aviation Fuel (SAF).

Electrochemical CO₂ Reduction: Uses electricity and single-atom catalysts to directly split CO₂ molecules, assembling them into chemical building blocks like polycarbonates and olefins used for making plastics.

4. Remove (Direct Extraction & Storage):

Direct Air Capture (DAC): Employs giant, fan-powered systems with liquid amine solvents or solid chemical sorbents to pull diffuse CO₂ straight out of ambient air.

Biomass-based approaches. Pyrolysis of converts biomass carbon into a stable form, Biochar, which can be sequestered in ground (agriculture fields, pits caused by soil extraction, etc) for permanent storage of carbon dioxide.

Bioenergy with Carbon Capture and Storage (BECCS): Burns fast-growing biomass (like wood pellets or agricultural waste) for electricity, then captures and buries the resulting emissions underground.

Enhanced Rock Weathering (ERW): Spreads finely crushed silicate rocks (like basalt) onto agricultural soils. The rock chemically reacts with rainwater to pull CO₂ from the atmosphere, turning it into stable bicarbonate minerals

The value of removal depends on permanence. Temporary storage, long-lived products and geological storage are not equivalent. Transparent standards, independent verification and monitoring systems are critical to credibility.

CARBON CAPTURE UTILIZATION & STORAGE (CCUS): represents a transformative approach with the capacity to create closed loop systems where CO₂ emissions can be captured, stored or converted into value-added products, thereby reducing the dependency on fossil fuels and minimizing waste.

CCUS technologies within the context of circular economy, encompass diverse carbon sources, viz fossil-fuel derived carbon, biomass and atmospheric. Multiple carbon capture technologies are available, with varying technology readiness level and economic viability. Likewise, various carbon storage options in geological formations, sequestration in agriculture fields and forests, along with direct and indirect utilization routes.

Policies, such as carbon tax, carbon markets and fiscal enablers favouring the implementation of CCUS into the circular economy are essential for commercialization of the technologies, proven at Lab or Pilot Plants stages. There is need for significant technology advancements to deploy these technologies at scale, as required to meet net-zero emissions targets,

Moreover, consolidated frameworks for assessment and monitoring, policy support, and international cooperation hold the potential to advance the adoption of this technology within a circular economy.

SHARED FRAMEWORK: The Circular Carbon Economy provides a shared framework through which these pathways can be aligned, measured and scaled. It supports national priorities such as energy security, industrial growth and rural development, while enabling international cooperation.

Opportunities include shared methodologies, common standards, joint research, demonstration projects and investment platforms and the development of markets for verified low-carbon products